Supplementary Material for "Respiratory-Phase Multipath Nulls in mmWave Cardiac Monitoring"

XinHao Yu

This supplement collects the tables and figures that support the paper without entering its argument. Tables S1–S5 give the evidence map, the Q_D regression, the breath-hold decomposition, the five-question summary and the effect of static-centre removal. Figures S1–S5 give the paired breath-hold rates, the drift-window sensitivity, the capsule phase portrait, a three-bandwidth example and the alignment distributions. The paper cites this material as supplementary material and, in one place, as supplementary Table S2.

TABLE S1
EVIDENCE BLOCKS AND THEIR ROLES

Dataset / block	N	Range resolution	Frame rate	Role
Numerical grid	143 combinations	—	—	failure region, D calibration
Lei dual-subject [18]	9 positions $\times$ 6 repeats $\times$ 3 bands (161 records)	5.0 / 6.0 / 7.5 cm	20 Hz	resolution control, D, geometry, frequency
110-subject [16]	110 subjects / 440 segments	31.25 cm	10 Hz	breath-hold causality only
Capsule [19]	20 records	not released	100 Hz	geometry, phase, Q_D
Emotion [17]	5 subjects / 38 trials	4.5 cm	100 Hz	cross-hardware geometry and phase

Bracketed numbers refer to the reference list of the paper. Rows are ordered by the role they play in the argument, not by size.

TABLE S2
Q_D REGRESSION ON THE 20-RECORD CAPSULE BENCHMARK

Metric	Pre-change	Q_D off	Q_D on
Field-level diffs.	—	0	355
Center-bin changes	—	0/20	0/20
Mean HR err (bpm)	6.01	6.01	5.63
Median HR err (bpm)	4.59	4.59	4.02
Per-case changes	—	—	2/20
Beat-level F1	0.5356	0.5356	0.5322
IBI RMSE (ms)	228.2	228.2	237.0

Both columns are paired against the same records; only the penalty is switched.

TABLE S3
BREATH-HOLD CONDITION DECOMPOSITION (A) AND PAIRED RATES IN THE TWO RESTING CONDITIONS (B)

(a) Condition	Median dips/min
Sitting, rest	20.4
Lying, rest	10.9
Sitting, post-exercise	34.8
Lying, post-exercise	22.4

(b) Rest	Inside	Outside	Ratio	Lower	p
Sitting	5.1/min	22.1/min	0.17	86%	2.0×10^{-11}
Lying	0.0/min	12.1/min	0.00	62%	1.4×10^{-4}

"Lower" is the fraction of subjects whose deep-dip rate falls inside the breath-hold window; p is a two-sided signed-rank test on the paired rates.

TABLE S4
FIVE-QUESTION SUMMARY

Question	Evidence	Answer
Does range resolution control the collapses?	5.0/6.0/7.5 cm, 161 records: 13.0/48.1/62.3%; $\lvert A_s\rvert/R$ 19.7/4.65/2.57; Spearman -0.56 to -0.81	Yes, through proximity to $\rho \approx 1$
Is respiration necessary?	Breath-hold, 110 subjects, rate ratio 0.44; dip frequency 92% (emotion), 0.00 median error (Lei, 6 cm)	Yes
Are the collapses geometric nulls?	Alignment gate $\kappa \geq 0.6$: 86–92% of Lei nulls, 78% capsule, 68% emotion; phase-step shape test against two synthetic worlds	A majority, but not all
Is the null at a fixed respiratory phase?	7/7 capsule; 13/25 emotion	Platform-dependent; the rest need a drifting reference
Is the failure region calibrated?	143-combination noisy grid, r–D bridge, 1450 dips	Yes: grid box $\rho \in [0.98, 1.15]$, $\theta_{\max} \geq 2.83$ rad; analytic band $\rho \in [0.74, 1.35]$ once $\theta_{\max} \geq \pi$

The bandwidth row and the grid row are the two ends of the argument: the first establishes that resolution moves the failure rate, the last that the failure region it moves across has a closed form.

TABLE S5
CENTRE REMOVAL DOES NOT REDUCE THE NULL RATE

Bin	None	Cplx. mean	Mov. mean	Geom. fit	Window. fit
5.0 cm	14.8%	96.3%	96.3%	53.7%	94.4%
6.0 cm	42.6%	88.9%	88.9%	70.4%	75.9%
7.5 cm	62.3%	88.7%	92.5%	71.7%	79.2%
All	39.8%	91.3%	92.5%	65.2%	83.2%

Record-level null rate ($n_{\text{deep}} \geq 5$ in 30 s) before and after removing an estimated static centre, on the same 161 records and under the same 15% depth criterion. No estimator lowers the rate: the mildest leaves it at 65.2% and the other four exceed 75%.

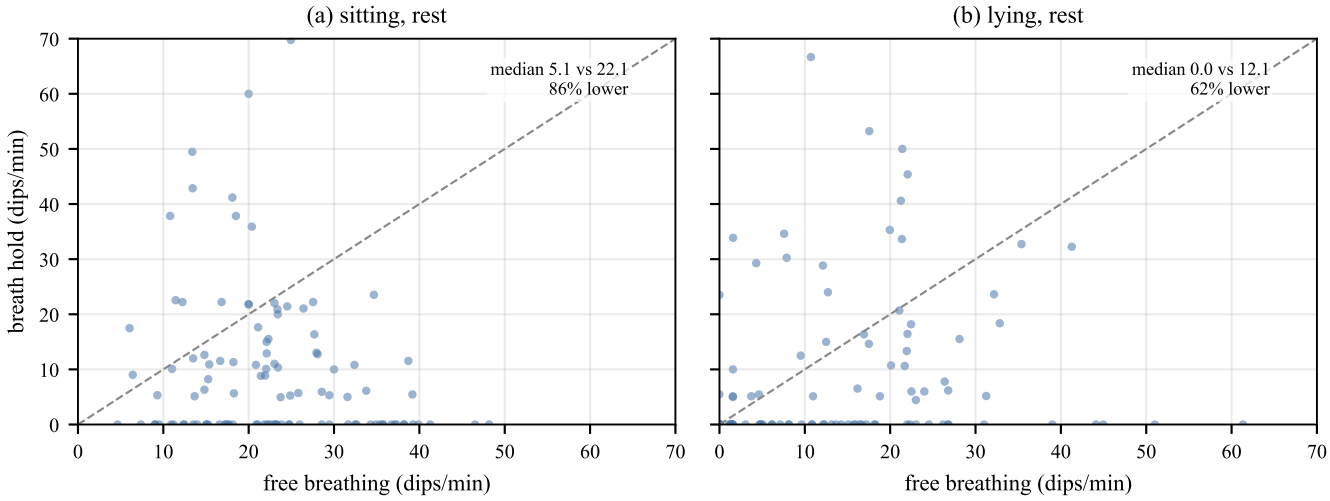


Fig. S1. Breath-hold control in the 110-subject dataset. Paired deep-dip rates inside versus outside the breath-hold window for (a) sitting and (b) lying rest. Dashed line: equality; medians and the fraction of subjects lower inside are annotated. The decomposition behind these panels is Table S3.

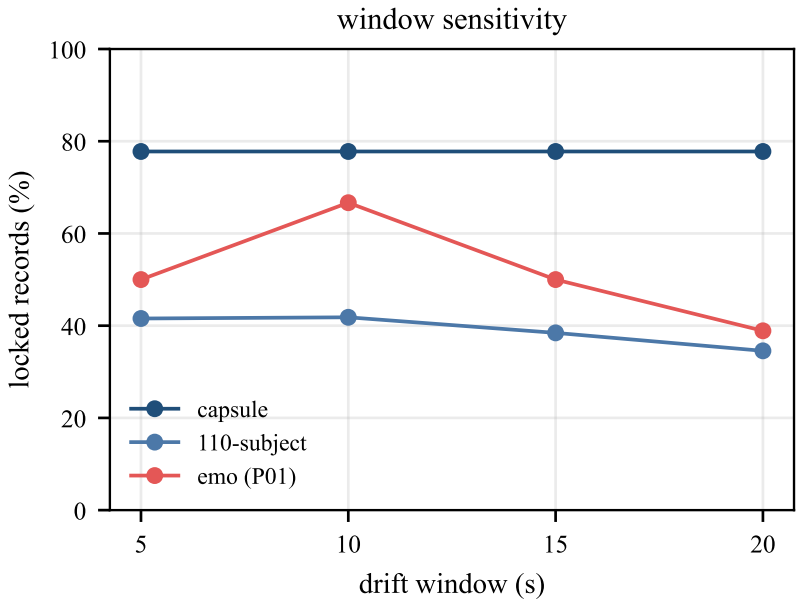


Fig. S2. Drift-window sensitivity. Locked fraction versus window length for the capsule, the 110-subject dataset and the emotion P01 subset. Ten seconds is the stable choice; 20 s over-smooths the respiratory motion.

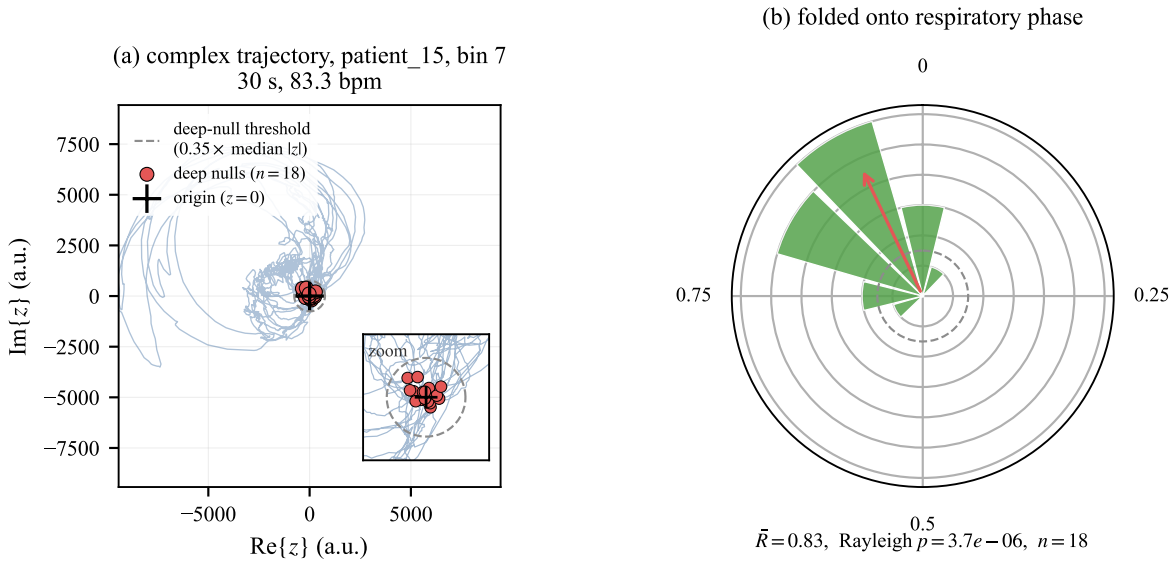


Fig. S3. Capsule representative record (patient_15, bin 7, 30 s, 83.3 bpm). (a) Complex trajectory; dashed circle: deep-null threshold; red dots: deep nulls ($n = 18$); cross: origin; inset: near-origin zoom. (b) Null instants folded onto the respiratory phase; $\bar{R} = 0.83$, Rayleigh $p = 3.7 \times 10^{-6}$.

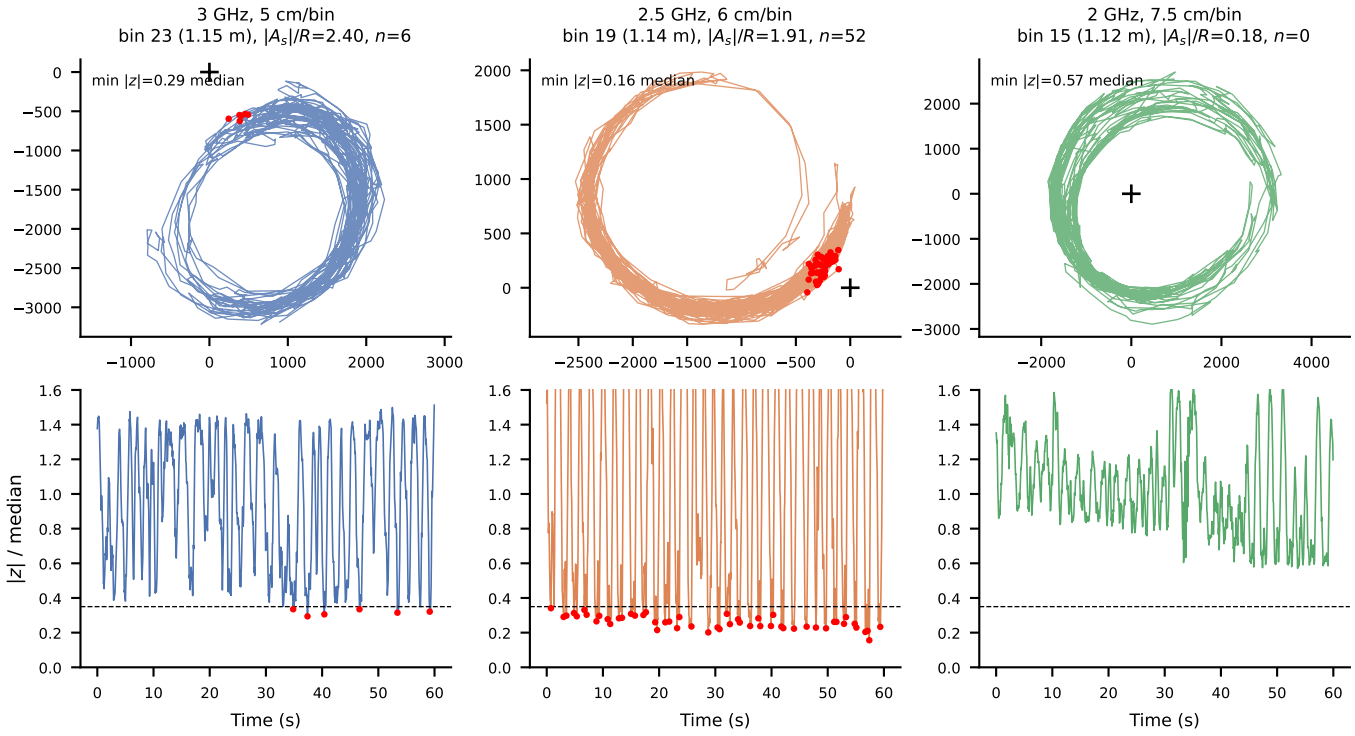


Fig. S4. Same position, three bandwidths (position 2, repeat 2; both subjects at 1.4 m). Selected bins at 1.15, 1.14 and 1.12 m, with $|A_s|/R = 2.40$, 1.91 and 0.18. Top: complex trajectory of the selected bin; cross: origin; red: deep nulls. Bottom: normalized envelope; dashed: $0.35\times$ median. The 6 cm bin repeatedly crosses the origin (52 nulls); the 7.5 cm bin is motion-dominated and never approaches it; the 5 cm bin grazes the origin and yields six shallow dips.

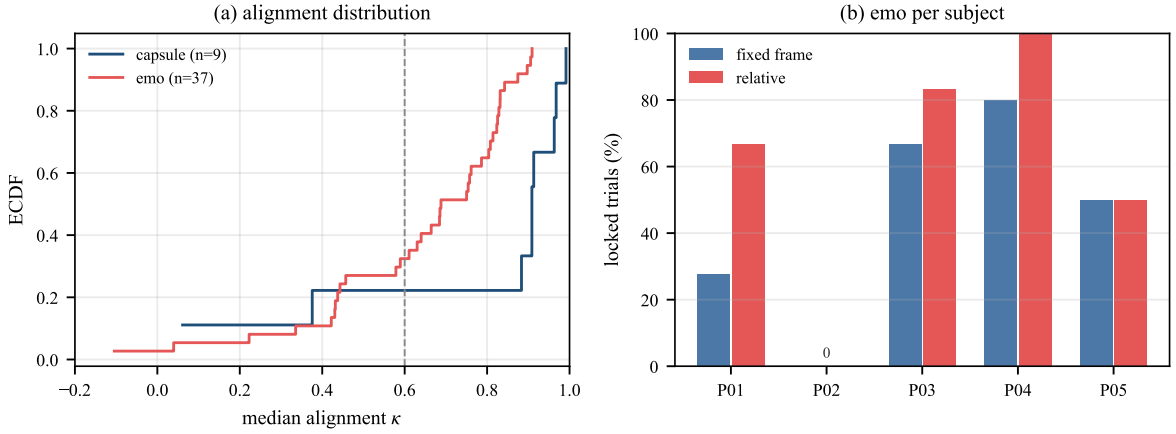


Fig. S5. Geometric classification and phase locking. (a) Empirical cumulative distribution of the median alignment κ for capsule ($n = 9$) and emotion ($n = 37$) records; dashed line: $\kappa = 0.6$ gate. (b) Locked-trial fraction per emotion subject for the fixed-frame and relative-phase analyses; P02 is zero in both.

Acknowledgment

The technical content of this work—the formulation, the derivations, the experiments, the analysis, and every numerical result reported in the paper and in this supplement—is the author's own. A large language model was used to translate and edit the English text. Its role was confined to wording: it did not generate or alter any measurement, figure, table, statistic, or conclusion. The author has read and verified the full text and takes responsibility for its accuracy.